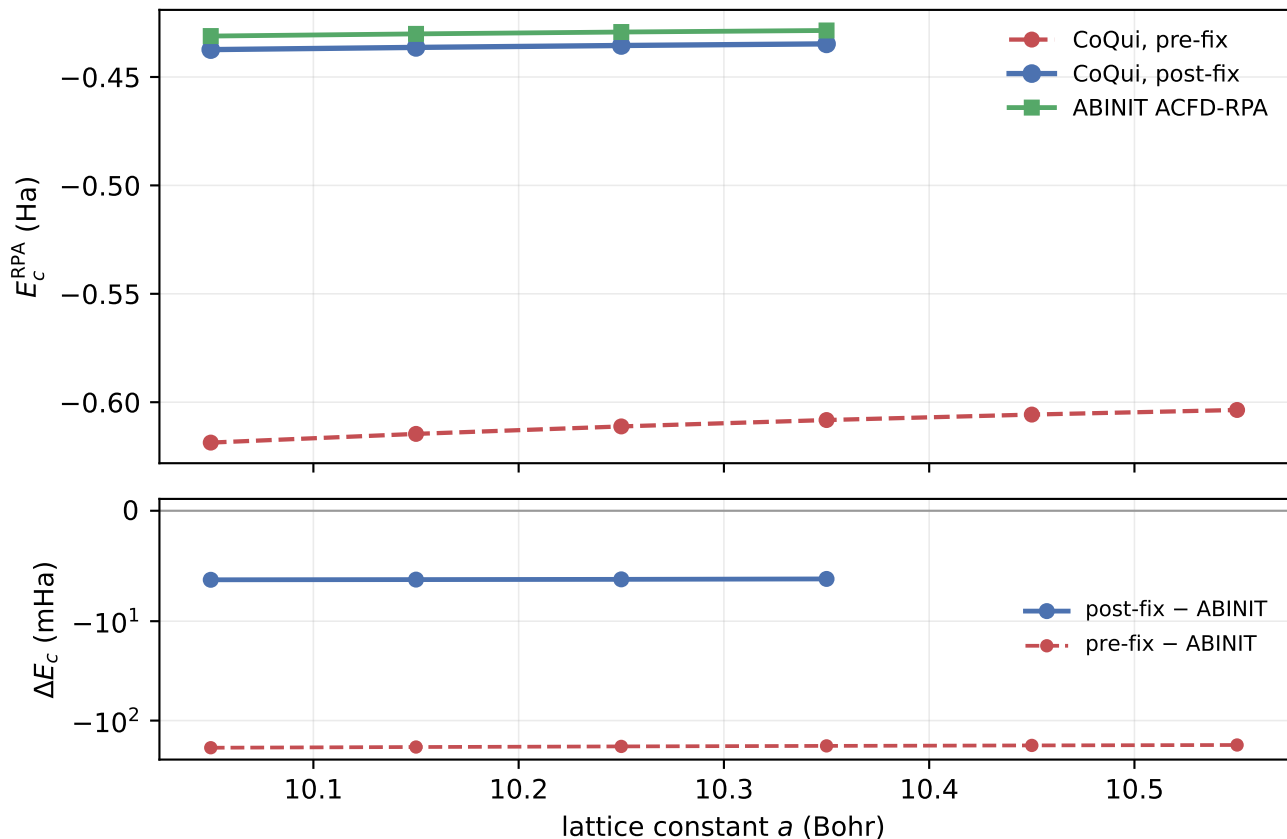


# Si RPA correlation energy, $n_{\text{band}} = 500$ , identical WFK



post-fix offset varies 0.08 mHa across the sampled range; pre-fix varies 11.5 mHa.  
 A CONSTANT offset cancels from the EOS; a volume-dependent one does not — that is why the pre-fix curvature ( $B_0 = 45$  GPa vs 98-101 reference) was destroyed while  $a_0$  looked fine.

Lower panel on a symlog axis (linear within  $\pm 10$  mHa). Both codes consume the bit-identical 500-band WFK, so  $E_c$  is the only difference.  
 Caveat: the pre-fix series also used looser thresh/tol, so it is not a pure  $V_{LL}$  contrast — but the compression defect it carries is itself volume-dependent, which is the same failure mode.